

SILICONA

INVESTOR BRIEF

AI-native semiconductor engineering, presented through measurable engineering evidence

Prepared for
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Classification
CONFIDENTIAL - INVESTOR DISCUSSION DRAFT

SILICONA is positioned around an evidence system for AI-assisted semiconductor engineering: AI proposes, deterministic EDA tools calculate, verification decides, and engineering approval governs progression.

AI proposes. EDA calculates. Verification decides. Engineer approves.

Source basis

This document is derived from the supplied SILICONA Independent Evaluation Report dated 30 September 2026 and the accompanying corrective review. Figures and statuses are reproduced as supplied. Claims are separated by evidence state, and unsupported superlatives or inferred competitor capabilities are excluded.

1. Investment thesis

SILICONA addresses a credibility problem inside AI-assisted hardware engineering: performance claims are easy to publish, while reproducible engineering evidence is expensive to generate and retain.

The supplied technical record describes a workflow linking AI proposals to deterministic EDA execution, certification gates, artifacts, hashes, benchmark results, and recorded failures. The business opportunity is to turn this evidence layer into infrastructure used by engineers, model providers, chip teams, and organizations evaluating AI-for-hardware systems.

The near-term commercial wedge is engineering and certification services. Paid model evaluation workloads follow. Enterprise platform subscriptions require operational scale. Evaluation data licensing is a longer-term opportunity dependent on corpus growth.

2. What exists today

The current record reports a functioning workflow rather than a slide-only concept. Its strongest evidence is concentrated in open-PDK implementation, certification, artifact provenance, and measured search workloads.

Evidence area	Reported state
Hash-bound artifacts	1,138
Recorded stage runs	563
Flow failures classified	104
Stage pass rate	81.0%
AI model evaluations	177
Current measured AI provider	GLM-4-Plus
Certified ladder levels	5 of 10, L1/L2/L3/L5/L9
Reference physical implementation	14 defined stages, DRC 0, dual LVS MATCH
Physical corpus	24 revisions
Largest reported search result	matrix-mac multi, n=5, +98.21%

3. Why the evidence matters

SILICONA's core distinction is not a claim that one AI model is better than another. It is the attempt to retain the engineering chain required to check a claim.

The proposed Evidence Graph links proposal, design, execution, failure, repair, verification, physical implementation, artifact, hash, benchmark result, and historical comparison. This creates a longitudinal record rather than a single demonstration.

The business consequence is a trust advantage for buyers who need to distinguish model output from engineering proof. The term Trust Asymmetry is used here as the consequence, not the underlying asset.

4. Technical evidence snapshot

Metric	Reported result	Interpretation
Defined pipeline	14 grouped stages	Explicit grouping resolves prior count ambiguity
Physical implementation	DRC 0, dual LVS MATCH	Open-PDK physical verification result
Timing	+5.74 ns pre-route / +3.48 ns post-route	OpenSTA result in supplied record
Fmax	240 MHz	Reported result for reference workload

Metric	Reported result	Interpretation
GDS	2,361,570 bytes	Artifact-bound output
Revisions	24	Reproducibility corpus
L8 audit	3 equivalence errors blocked certification	Governance evidence
AI evaluations	177	All current corpus on GLM-4-Plus
Search results	+98.21%, +79.65% average, +34.30%	Measured families with small sample sizes

5. Product architecture

The operating model follows four governing responsibilities.

Responsibility	System role
AI proposes	Generate or refine engineering candidates, specifications, RTL, search candidates, and repair proposals
EDA calculates	Run deterministic synthesis, simulation, formal verification, equivalence, physical implementation, timing, DRC, LVS, and related tooling
Verification decides	Gate certification states using deterministic tool verdicts and recorded evidence
Engineer approves	Retain policy authority over bounded automation and certification progression

6. Certification ladder and roadmap

The current ladder reports five certified levels. The remaining levels are represented as templates or measured states pending qualifying evidence.

Current level	Status	Next evidence objective
L1	CERTIFIED	Maintain corpus
L2	CERTIFIED	Maintain formal proof corpus
L3	CERTIFIED	Maintain protocol IP evidence
L4	TEMPLATE	Qualifying ALU run
L5	CERTIFIED	Maintain subsystem evidence
L6	TEMPLATE	Qualifying RV32I run
L7	CERTIFIED	Expand accelerator search evidence
L8	TEMPLATE	Resolve complete-SoC equivalence requirements
L9	CERTIFIED	Expand physical implementation corpus
L10	MEASURED	Increase sample sizes and baseline coverage

7. Market position

The supplied competitive material points to a gap around independently checkable AI-for-EDA performance claims. The appropriate comparison is therefore about published evidence practices, not assumptions about competitors' private engineering results.

The supplied report references Synopsys, Cadence, Rrecursive Intelligence, ChipAgents, Agentrys, Cognichip, Silimate, MooresLabAI, and CaretEDA. Their published claims, funding references, and disclosed protocols should be treated as external-source records, not as evidence of their internal product state.

SILICONA's stated position is evidence infrastructure for AI-designed and AI-assisted silicon. The opportunity extends across engineering teams, AI model providers, chip startups, EDA workflows, and organizations requiring independent evaluation.

8. Business model

The revenue sequence should follow dependency and customer readiness.

Phase	Revenue stream	Why it comes first
Today	Engineering and certification services	Direct service sale with limited enterprise dependency
Early customer phase	Paid evaluation workloads, Model Lab	Monetizes independent model measurement as corpus grows
Scale phase	Enterprise platform subscriptions	Requires hosted stability, support, and distributed compute
Strategic phase	Benchmark and evaluation data licensing	Requires sufficient volume and longitudinal quality

9. Service wedge

The supplied commercial plan describes evidence-bound certification packs for shuttle submissions and related engineering workflows, with a chipIgnite target of \$14,950. This figure is a proposed service price, not reported revenue.

The service model also feeds the Evidence Graph. Each paid workload produces additional engineering evidence, failure classifications, and artifact history, subject to customer confidentiality and data-rights terms.

10. Infrastructure requirement

The present development environment is approximately 4 GB RAM, 10 GB disk, and 2 cores. The production BOM specifies substantially larger solver, physical-design, GPU, storage, and network resources.

The year-one planning envelope in the supplied BOM is approximately \$150,000. Before external compute is activated, the technical review identifies a global concurrency limiter, disk-space guard, artifact blob storage, hardware-class timeouts, a multi-node database decision, and automated scratch tiering as prerequisites.

Resource	Current	Production planning target
RAM	~4 GB	256 GB+ solver fleet; 512 GB to 1 TB physical design
CPU	2 cores	64+ formal; 32+ synthesis
GPU	0	H200-class inference and larger accelerator pool
Storage	~10 GB	TB-scale NVMe and object storage
Network	Sandbox	25 to 100 Gbps physical-design I/O
Year-one BOM	Not provisioned	~\$150K planning envelope

11. What the financing would fund

The evidence supports an infrastructure-led next phase rather than a purely promotional expansion. Capital or cloud credits would primarily fund reproducible execution at higher scale, cross-provider measurement, physical-design workloads, artifact storage, and the engineering controls required for distributed operation.

A separate legal and IP structure is also required for institutional financing readiness. The supplied review identifies incorporation and IP assignment as immediate structural work.

12. Indicative financing scenarios

The figures below are scenarios from the supplied evaluation, not independently established valuations. They should be treated as planning ranges rather than assertions of current market value.

Scenario	Indicative figure	Stated catalyst or condition
Pre-seed / IP value scenario	\$3M to \$6M	Working codebase, defined pipeline, artifacts, Benchmark Protocol
Seed post-money scenario	\$12M to \$18M	Incorporation, IP assignment, Phase A benchmark completion
12 to 18 month scenario	\$25M to \$45M	Cloud partnership, open-MPW shuttle result, Phase B data
2 to 3 year scenario	\$80M to \$150M	Revenue, Compute Federation, multiple measured providers, enterprise pilots
5+ year scenario	\$500M to \$1.5B+	Industry-standard evidence infrastructure or strategic acquisition scenario

These scenarios depend on corporate structure, market conditions, evidence quality, revenue, customer traction, infrastructure access, investor demand, and negotiation. They are not transaction comparables or independently verified valuations.

13. Milestones that change the evidence profile

Phase A: complete search-family coverage at $n \geq 5$ with recorded seeds and publish the methodology.

Phase B: execute cross-provider measurement across the planned model set, including OpenAI, Anthropic, Google, DeepSeek, Mistral, Qwen, and GLM.

Phase C: achieve independent third-party reproduction of selected runs using a published manifest.

Physical milestone: submit an initial open-MPW shuttle design using the certified flow and pursue physical silicon.

Commercial milestone: convert the certification service into paid customer workloads and document repeatable delivery economics.

Infrastructure milestone: deploy Compute Federation with concurrency, storage, timeout, and state controls in place.

14. Risk register

The principal risks are measurable and therefore suitable for milestone tracking.

Risk	Current condition	Response
Single-provider AI corpus	177 evaluations, 100% GLM-4-Plus	Cross-provider Phase B
Compute ceiling	Current environment materially below production BOM	Cloud and infrastructure partnerships
Legal structure	Unregistered status identified in supplied review	Establish entity and assign IP
Founder concentration	High dependency on founder execution	Document, automate, hire targeted roles
Certification trust	L8 false claim was blocked by equivalence failures	Maintain deterministic certification state machine
Distributed execution	Resource storms remain a concern	Complete Phase 0 controls first
Export-control exposure	Jurisdiction-dependent	Portable routing subject to applicable requirements

15. Competitive evidence discipline

SILICONA should avoid claims about what competitors have or have not achieved internally unless those facts are directly documented by reliable public evidence.

The investor-facing comparison should instead show SILICONA's own protocol, evidence levels, artifact chain, failure retention, and reproducibility status. External company references should be dated and attributed to

their published materials.

16. Why the next phase is measurable

The roadmap is unusually amenable to objective milestones. More compute produces larger workloads. More providers produce cross-provider measurements. More revisions produce a larger evidence graph. Independent reproduction tests the portability of the evidence. Physical silicon tests the transition from open-PDK implementation to manufactured hardware.

The result is a sequence of evidence upgrades rather than a sequence of narrative upgrades.

17. Investor diligence package

The Technical Evidence Dossier should accompany this brief. A serious diligence package should also include the Benchmark Protocol, artifact manifest, toolchain versions, selected run logs, failure classifications, infrastructure BOM, corporate and IP documents when available, and a reproducibility manifest.

The supplied evidence does not establish E4 independent reproduction, manufactured silicon, recurring revenue, or broad enterprise adoption. Those items belong in the milestone plan rather than in the present evidence base.

18. Closing position

SILICONA's current proposition is strongest when expressed without superlatives: a traceable engineering system connecting AI proposals to deterministic EDA outcomes, certification gates, artifacts, benchmarks, and failures.

The immediate objective is to widen the evidence base and move selected results from artifact-bound evidence toward independent reproduction. In parallel, the service wedge provides a route to early revenue while the platform matures.

The Technical Evidence Dossier contains the detailed record. This brief provides the financing and commercial context around it.

AI proposes. EDA calculates. Verification decides. Engineer approves.